Derick Conrad D. Mercado

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EDUCATION & ORGANIZATIONS

De La Salle University - Dasmarinas – BS Computer Science | GPA: 4.00/4.00 *2024 – Present*

Relevant Coursework: Software Engineering, Data Structures & Algorithms, Database Systems, Artificial Intelligence, Machine Learning, Natural Language Processing, Operating Systems, Statistics, Discrete Mathematics

Awards: Dean's Lister SY 2024-2026, Top 1 Overall General Population Academic Scholar

College of Information and Computer Studies Student Government – *Production & Operations Committee* *2025 – Present*

Computer Science Program Council – *Campaign Committee* *2025 – Present*

WORK EXPERIENCE

Freelance Database Analyst - *EEI CREATIVE INTERNATIONAL PTE. LTD.* *Sept 2024 – Jan 2025*

- **Built a Python CLI web scraper** with requests and BeautifulSoup4 that pulls emails, phone numbers, addresses, and social profiles across **7 platforms** from any website in **under 10 seconds**, with parsing and extraction done in **under 1ms per page.**
- **Increased data yield by 4x** by adding multi-page crawling that automatically follows up to **3** contact-related sub-pages per site in a single command, also **supports multiple URLs** in one command.
- **Reduced false-positive output by 90%+** by implementing validation rules for malformed emails, short/invalid phone numbers, and irrelevant links, with each data type handled in its own regex module.

PROJECTS

MediQueue – *Lead Developer | Academic Project*

- **Streamlined patient onboarding workflows** by engineering a **6-state** hospital queue management platform with atomic SQL transitions, **REST API integrations** using React, Express (TypeScript), and PostgreSQL.
- **Cut API latency by 52%** by designing a FIFO queue engine on PostgreSQL with connection pooling and Supabase JWT auth, **dropping response time from 389ms to 185ms** and **sustaining 43.6 req/sec at 100% success rate** across 200 load tests.
- Eliminated unauthorized access by implementing JWT role-based access control with admin-only mutations and auto-logout, sustaining **100% success rate** across 200 authenticated load-test requests under 10 concurrent users.
- Delivered **live queue position updates in under 10 seconds** by implementing client-side polling with change detection, sustaining **689ms avg response time** at **100% success rate** across 50 requests under 10 concurrent users.
- Achieved **production-ready, single-command deployment** by containerizing the full stack with Docker multi-stage builds, Kubernetes health probes, resource limits, and RollingUpdate strategies across 2 replicas per service with a PowerShell script.

LogiBot – *Personal Project*

- Diagnosed medical symptoms with **1.93ms avg response time** by building a graph-based engine mapping cross-body symptoms via directed edges and weighted urgency scoring, letting users click anatomical regions on an interactive 3D human model to receive results — **sustaining 789.5 req/sec** at **100% success rate** across 700 load tests under 50 concurrent users.
- Identified conditions in **as few as 2 turns** by implementing a multi-turn conversational state machine with server-side session accumulation and **zero external ML dependencies**.

CERTIFICATIONS

Harvard CS50 *June 2025*

ACTIVITIES

SIKAPTala Hackathon 2026 – *Top 14 out of 61* *May 2026*
Microsoft's Fabric + Foundry Workshop *Nov 2025*
Global AI Manila MCP Workshop *Dec 2025*

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks & Databases: React, Node.js, Express.js, Flask, Tailwind, Bootstrap, Supabase, MySQL, SQLite, PyTorch, Tensorflow
Developer Tools: Github, Git, VS Code, Docker Desktop, Kubernetes, Google Colab, MATLAB